Tricky Topic

Factorising quadratics

You need to be able to **factorise** quadratic expressions by writing them as the product of **two brackets**. Make sure you are really confident expanding **double brackets** before you start. You can revise this on page 43. You can also revise factorising simpler expressions on page 29.

Expanding brackets →
$(x + 2)(x + 3) = x^2 + 5x + 6$
← Factorising

Factorising $x^2 + bx + c$

You need to write the expression with **two brackets**. Look for two numbers whose **sum** is equal to b and whose **product** is equal to c.

You need to find two numbers which add up to 7... $5 + 2 = 7$

$x^2 + 7x + 10 = (x + 5)(x + 2)$

... and multiply to make 10 $5 \times 2 = 10$

You can use this table to help you find the two numbers.

b	c	Factors
positive	positive	both numbers positive
positive	negative	bigger number positive and smaller number negative
negative	negative	bigger number negative and smaller number positive
negative	positive	both numbers negative

Worked example

Factorise $x^2 - x - 20$ **(2 marks)**

Factor pairs of 20:
1 and 20, 2 and 10, 4 and 5

$x^2 - x - 20 = (x + 4)(x - 5)$

Check:

$(x + 4)(x - 5) = x^2 - 5x + 4x - 20$
$= x^2 - x - 20$ ✓

Examiners' report

The answer will have **two sets of brackets**.
The last term is negative, so the brackets will have one + sign and one − sign: $(x + \square)(x - \square)$.
The numbers will be a **factor pair of 20**.
With any factorisation, the safest thing to do is to **check your answer** by expanding the brackets.

Real students have struggled with questions like this in recent exams – **be prepared!**

Difference of two squares

You can factorise expressions that are written as

(something)^2 − (something else)^2

Use this rule:

$a^2 - b^2 = (a + b)(a - b)$

LEARN IT!

$x^2 - 36 = x^2 - 6^2$
$= (x + 6)(x - 6)$

36 is a square number.
$36 = 6^2$ so $a = x$ and $b = 6$

Worked example

Factorise $x^2 - 9$ **(2 marks)**

$x^2 - 9 = (x + 3)(x - 3)$

Check:

$(x + 3)(x - 3) = x^2 - 3x + 3x - 9$
$= x^2 - 9$ ✓

Keep an eye out for the difference of two squares. $9 = 3^2$ so $a = x$ and $b = 3$.

Now try this

1 Factorise
(a) $x^2 + 6x + 8$ **(2 marks)**
(b) $x^2 - 10x + 16$ **(2 marks)**

Worked solution video

2 Factorise
(a) $x^2 - 144$ **(2 marks)**
(b) $x^2 - 49$ **(2 marks)**

Tricky Topic

Quadratic equations

You need to be able to **solve** a quadratic equation **without a calculator** by factorising one side. If you need to solve a quadratic equation in your Foundation GCSE exam it will look something like this:

The left-hand side will be a **quadratic expression.**

$$x^2 + 6x + 5 = 0$$

The right-hand side will be **zero.**

Follow the steps below to solve the equation.

1 Factorise the left-hand side of the equation:
$(x + 5)(x + 1) = 0$

For a reminder about factorising quadratic expressions look at page 46.

2 Set each factor equal to zero:
$(x + 5)(x + 1) = 0$
$(x + 5) = 0$ $(x + 1) = 0$

3 Solve the two linear equations to find the two solutions:
$x + 5 = 0$ $x + 1 = 0$
$x = -5$ $x = -1$

Write both solutions in the form $x =$ ☐

Worked example

Solve $x^2 + 8x - 9 = 0$ **(3 marks)**

$(x + 9)(x - 1) = 0$

$x + 9 = 0$ $x - 1 = 0$

$x = -9$ $x = 1$

1. Start by factorising the left-hand side of the equation. You can check by expanding the brackets:
$(x + 9)(x - 1) = x^2 - x + 9x - 9$
$= x^2 + 8x - 9$ ✓
2. Set each factor equal to zero.
3. Solve the linear equations to get **two** solutions.

Check it!
$(1)^2 + 8(1) - 9 = 1 + 8 - 9 = 0$ ✓
$(-9)^2 + 8(-9) - 9 = 81 - 72 - 9 = 0$ ✓

Worked example

Solve $x^2 - 100 = 0$ **(3 marks)**

$(x + 10)(x - 10) = 0$

$x + 10 = 0$ $x - 10 = 0$

$x = -10$ $x = 10$

You can also use inverse operations:
$x^2 - 100 = 0$ $(+ 100)$
$x^2 = 100$ $(\sqrt{\ })$
$x = 10$ or $x = -10$
There are two answers because
$(-10)^2 = -10 \times -10 = 100$

Two to watch

1 When one solution is $x = 0$
$x^2 - 10x = 0$
$x(x - 10) = 0$
Solutions are $x = 0$ and $x = 10$

2 Difference of two squares
$x^2 - 25 = 0$
$(x - 5)(x + 5) = 0$
Solutions are $x = 5$ and $x = -5$

Now try this

For parts (a) and (c) look at the blue box above.

1 (a) Factorise $x^2 + 5x + 6$ **(2 marks)**
(b) Write down the two solutions of
$x^2 + 5x + 6 = 0$ **(1 mark)**

Both solutions are **negative.**

2 Solve
(a) $x^2 - 5x = 0$ **(2 marks)**
(b) $x^2 + 3x - 28 = 0$ **(3 marks)**
(c) $x^2 - 144 = 0$ **(3 marks)**

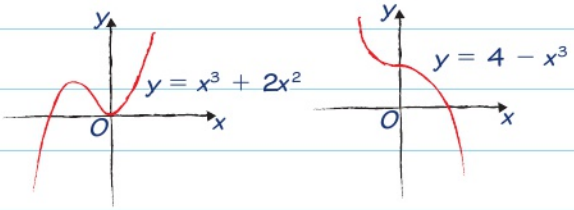
Tricky Topic

Cubic and reciprocal graphs

You might need to **draw** or **interpret** cubic and reciprocal graphs in your exam. You can use a **table of values** to draw any graph, but it helps if you know what the general shape of the graph is going to be.

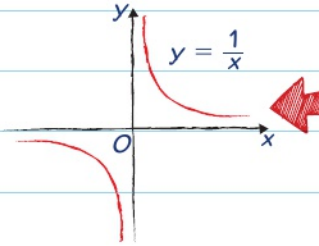
1 Cubic graphs

Graphs that contain an x^3 term and no higher powers of x are called cubic graphs. Here are two examples:



2 The reciprocal graph

The graph of $y = \frac{1}{x}$ is called the reciprocal graph.



The graph gets closer and closer to the x- and y-axes but never touches them.

Worked example

(a) Complete the table of values for $y = x^3 - 4x - 3$

x	−2	−1	0	1	2	3
y	−3	0	−3	−6	−3	12

(2 marks)

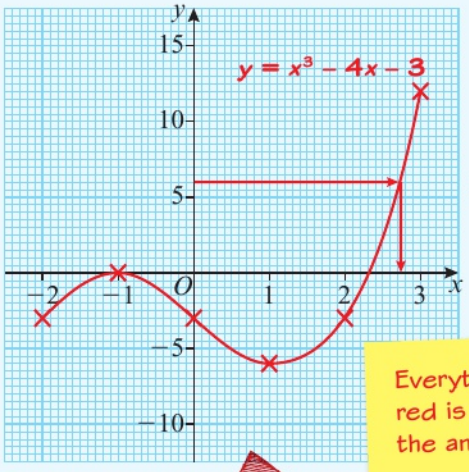
(b) On the grid, draw the graph of $y = x^3 - 4x - 3$ for $-2 \leqslant x \leqslant 3$ **(2 marks)**

(c) (i) Estimate the value of x when $y = 6$ **(1 mark)**

2.7

(ii) Comment on the accuracy of your estimate. **(1 mark)**

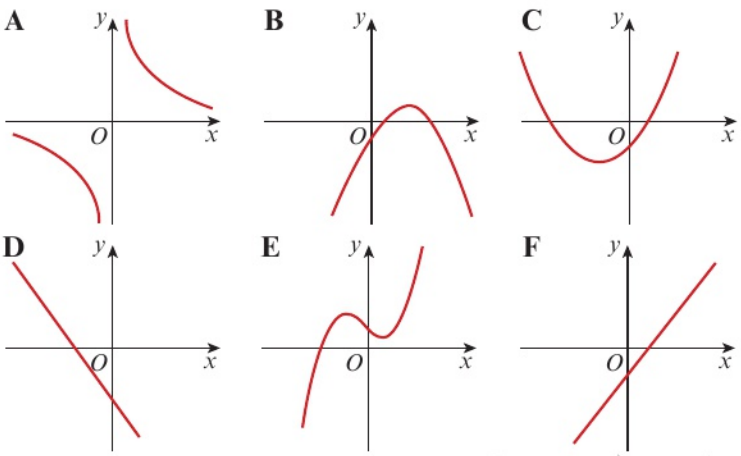
The estimate is not very accurate because it is based on reading off a graph.



Everything in red is part of the answer.

This is a **cubic graph** with a **positive** coefficient of x. If you recognise the shape of the graph then it's easier to tell if you've plotted your coordinates correctly.

Now try this



Write down the letter of the graph which could have the equation

(a) $y = -4x - 2$

(b) $y = x^3 - 3x + 4$

(c) $y = \frac{1}{x}$

(d) $y = 8x - 15 - x^2$

(e) $y = 2x - 3$

(f) $y = x^2 + 2x - 8$ **(6 marks)**